Project Design Book and Website (team) (Week 13)

Start Assignment

- Due 4 Dec by 23:59
- Points 53
- Submitting a file upload
- Available 19 Oct at 0:00 - 9 Dec at 23:59

Purpose

The purpose of this submission is to demonstrate that you and your team can develop an interactive data visualisation for the Data Visualisation Project that meets user needs.

The submission consists of two parts:

1. Design Book (evidence of data processing and your design process)
2. Website displaying an interactive data visualisation and associated code

Your Project must satisfy the brief provided in: **2026 Data Visualisation Project Brief (https://swinburne.instructure.com/courses/78028/pages/2026-data-visualisation-project-brief)**

This is a **team** deliverable worth **40%** of your final mark.

Teams will be formed by your tutor in Week 8.

Students will be required to work on their own if they:

- have NOT made an effort to be placed in a team **before** Week 9,
- have NOT meaningfully communicated with their team before Week 9 Class, **or**
- have NOT contributed to teamwork before Week 10 Stand-up.

Submission Requirements

File 1: Design Book

Format: Written Assignment (IMPORTANT - see: **Design Book Template (https://swinburne.instructure.com/courses/78028/pages/design-book-template)**)

Word Limit: None

Filetype: PDF

File 2: Data Processing

Format: knwf (KNIME file - file and WORKFLOW name MUST include team name, file **must include data**)

Links: Website

Item: Link to website hosted on COS30045 Mercury website with charts programmed in D3.

Item: Link to COS30045 GitHub site containing code.

Note: Code MUST be commented and GenAI acknowledged in comments where appropriate

Links MUST be posted in comment section of submission.

Academic Integrity

See: **Academic Integrity (https://swinburne.instructure.com/courses/78028/pages/academic-integrity)**

References must be in APA 7 format.

IN ADDITION - You must submit work in progress for each Stand-up.

That is, you must:

- submit expected milestones for Stand-ups,
- demonstrate understanding of work submitted for Stand-up

Students who achieve fail marks on a Project Stand-up may have their mark reduced relative to the achieved mark.

Lack of continuity between final Project and works submitted for Stand-ups may result in an integrity investigation and/or a reduction in your Project mark relative to achieved mark.

Teamwork

Where the project is completed in a team:

- Team members must complete the Peer Assessment
- Non-contribution to a team assessment is considered a matter of Academic Integrity.
- Each team member must contribute to, and be knowledgeable about, BOTH the design AND programming of the data visualisation

Students who are rated poorly by their team mates, or are otherwise identified as not fully participating in the Project, may have their mark reduced relative to the achieved mark.

Extensions and Late Penalties

See: **Extensions and Late Penalty Policy (https://swinburne.instructure.com/courses/78028/pages/extensions-and-late-penalty-policy)**

Rubrics

See below for rubrics that explain how this assessment will be marked. GenAI was used to refine the rubrics for this assessment.

27/5 Rubrics for Code implementation have been updated.
Please note the following important points:

Data Exploration Criteria: To pass Data Exploration criteria, KNIME file must be included and *contain all the data used in vis* (including exported files used in visualisation).

Coding Practice Criteria: To pass the Coding Practice criteria, code must be commented and must include summary of AI prompts. If you did not use AI you must state it in the code comments.

Design Process Book 6

Criteria	Ratings					Pts
Introduction and Purpose	**4 to >3.0 Pts** **Excellent** The introduction clearly defines the problem context and target audience, showing a strong understanding of who the visualisation is designed for and why. The motivation for the visualisation is well explained and appropriately grounded in the domain context. User tasks and key questions are clearly specified and closely aligned with the stated purpose of the visualisation. The benefits and intended impact of the visualisation are thoughtfully explained, particularly in relation to supporting analysis or decision-making. Relevant academic literature, domain research, or authoritative sources are used appropriately to support claims	**3 to >2.6 Pts** **Good** The introduction defines the target audience and explains the purpose and importance of the visualisation in relation to the context. User tasks and questions are identified and mostly aligned with the visualisation goals. The benefits and impact of the visualisation are explained with reasonable depth. Some reference to relevant literature or domain context is included, though integration of sources may be limited.	**2.6 to >2.2 Pts** **Satisfactory** The target audience and overall purpose are identified, but explanations may be broad or somewhat generic. The motivation for the visualisation is described but lacks depth or strong contextual grounding. User tasks, questions, or benefits are stated but may be incomplete or loosely connected. Little or no academic or domain literature is referenced.	**2.2 to >1.8 Pts** **Limited** The introduction provides a basic description of the visualisation and its audience, but key elements are vague, incomplete, or weakly justified. The importance of the visualisation and intended user tasks are mentioned with limited clarity or relevance. Benefits and impact are poorly articulated.	**1.8 to >0 Pts** **Unsatisfactory** Any of the following: The introduction does not clearly define the target audience, motivation, or purpose of the visualisation. User tasks and intended outcomes are missing or unclear, or there appears to be little connection to the problem context. Content is generic, vague, or contains few concrete examples. Lacks logical flow, content is very difficult to follow.	4 pts

Criteria	Ratings					Pts
	and demonstrate a deep understanding of the problem domain.					
Data Source and Governance	**3 to >2.25 Pts Excellent** The data sources are clearly identified, appropriately referenced, and relevant to the stated analytical questions. Data governance is acknowledged. The relationship between the data and the questions outlined in Section 1.2 is explicitly and convincingly explained.	**2.25 to >1.95 Pts Good** Data sources are clearly identified and relevant, with appropriate links and a reasonable data summary. Data governance is acknowledged. The connection between the data and the intended analytical questions is explained.	**1.95 to >1.35 Pts Satisfactory** Data sources are identified but documentation may be brief or incomplete. The data summary provides basic information but lacks detail. Data governance is acknowledged. The link between the data and the analytical questions is stated but not strongly justified.	**1.35 to >0 Pts Unsatisfactory** Any of the following: Data sources are not directly referenced in reference section. No link to data is provided or does not provide access to the data. It is not clear how the data supports the intended data visualisation questions.		3 pts
Data Processing and Exploration	**5 to >3.75 Pts Excellent** Data attributes and data types are correctly identified. The data cleaning, transformation, filtering, and integration steps are clearly described and well justified, with evidence of careful reasoning. The KNIME workflow is clearly presented and supports the documented process. Exploratory	**3.75 to >3.25 Pts Good** Key data attributes and data types are mostly correct. The data cleaning and processing steps are described clearly, with appropriate use of KNIME workflows. Exploratory analysis includes relevant summaries and screenshots of visualisations, and key observations are discussed, though reflection on limitations or	**3.25 to >2.75 Pts Satisfactory** Data processing steps are described but may lack clarity, justification, or completeness. Exploratory analysis is present but limited, with basic observations and minimal discussion of challenges or data quality issues. KNIME workflow submitted and contains all data sets used in analysis.	**2.75 to >2.25 Pts Limited** Data types, or cleaning processes is minimal or partially incorrect. The relationship between the data and the visualisation purpose is weakly articulated or implicit. Exploratory analysis is superficial, with limited evidence of meaningful engagement with the data. KNIME workflow submitted and	**2.25 to >0 Pts Unsatisfactory** Any of the following: Data processing and exploration are poorly documented or not described. ### KNIME workflow is missing or data is missing. ### Content is generic, vague, or contains few concrete examples. Lacks logical flow, content is difficult to follow.	5 pts

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	data analysis is systematic and informative, using appropriate summary statistics and visualisations. Initial observations are clearly discussed, including notable patterns, anomalies, and challenges, demonstrating a strong understanding of the data and its limitations. KNIME workflow submitted and contains all data sets used in analysis.	challenges may be limited. KNIME workflow submitted and contains all data sets used in analysis with appropriate annotations.		contains all data sets used in analysis. Full annotation of KNIME file may be missing.		
Website Design	**3 to >2.25 Pts Excellent** The wireframe clearly represents the intended webpage structure, including navigation, layout, and the placement of headings, text, and visual elements. The design demonstrates a strong understanding of user needs and supports the stated	**2.25 to >1.95 Pts Good** The wireframe clearly shows the main layout, navigation, and placement of content elements. The storyboard describes how users will interact with the website and visualisation in a generally clear and logical way. The proposed structure supports the visualisation's purpose, though	**1.95 to >1.65 Pts Satisfactory** The wireframe presents a basic page layout and content placement, but navigation or structure may be simplistic or partially unclear. The storyboard outlines user interaction at a high level but lacks detail or coherence in the user journey. Connections between the design and the intended purpose	**1.65 to >1.35 Pts Limited** A wireframe and storyboard are provided but are minimal, incomplete, or difficult to interpret. The layout and interaction flow show limited consideration of user needs or the stated purpose of the visualisation. Navigation and user flow may be unclear or poorly structured.	**1.35 to >0 Pts Unsatisfactory** The wireframe or storyboard is missing, unclear, or does not meaningfully represent the intended website structure or user interaction. There is little evidence of planning for layout, navigation, or how users would engage with the visualisation.	3 pts

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	purpose of the visualisation. The storyboard clearly and logically illustrates how users will navigate the site and interact with the visualisation to answer key questions or achieve the intended goal. Interactions and transitions are purposeful, well sequenced, and aligned with the narrative. The wireframe and storyboard together show thoughtful consideration of clarity, usability, and information flow.	some aspects of interaction flow or layout decisions may not be fully explained or optimised.	are present but weakly justified.		Content is generic, vague, or contains few concrete examples. Lacks logical flow, content is difficult to follow.	
Visualisation Design	**10 to >7.5 Pts Excellent** The choice of chart types are clearly explained and well justified in relation to specific user needs, data characteristics and design guidelines. Alternate designs are considered in some depth. A strong informed argument is made that	**7.5 to >6.5 Pts Good** The choice of chart types has been justified in relation to the data and purpose of the visualisation. The design generally follows good visualisation principles, with some consideration of accuracy and accessibility. Visual encoding (i.e., marks and	**6.5 to >5.0 Pts Satisfactory** Reasons for choosing chart types are provided and show good understanding of the data, visual encoding (e.g., marks and channels) and design principles. Alternate designs are not considered in any depth. Accessibility is discussed and an evaluation of colour and contrast are	**5 to >4.0 Pts Limited** Chart choices and design elements are described, but justifications are weak, unclear, or incomplete. Discussion of design principles is minimal. Accessibility may be mentioned but not formally addressed. Limited discussion of	**4 to >0 Pts Unsatisfactory** Any of the following: Chart choices and visual design decisions are unclear, inappropriate, or not explained. There is little evidence of consideration for visualisation principles, accessibility, or accurate data representation. Design elements do not	10 pts

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	chosen designs follow good visualisation principles, including graphical integrity and accessibility. Visual encodings (i.e., marks and channels) are demonstrated to have been used purposefully to represent data differences and relationships accurately. Colour palettes, labelling, and layout decisions are clearly justified and support clarity, comparability, and readability. The planned use of annotations showing how they enhance user understanding without clutter or distraction. Sketches and/or screenshots are used effectively to illustrate and support key design decisions.	channels) are mostly well described though some choices may not be fully justified. Colour, labelling, annotations and layout decisions are demonstrated to support understanding. Supporting sketches or screenshots are provided to illustrate design decisions but may be limited in detail.	provided. Some evidence of minor misunderstandings of visualisation design concepts. Sketches or screenshots are provided but offer limited insight into decision-making.	choice of visual encoding (i.e., marks and channels), colour, and layout. Some concepts appear to be misunderstood or misapplied. Supporting sketches or screenshots are minimal or difficult to interpret.	support user understanding or may mislead users, and supporting sketches or evidence of design planning are missing. Visual encoding (i.e., marks and channels) is not discussed or discussion demonstrates poor understanding. Content is generic, vague, or contains few concrete examples. Lacks logical flow, content is difficult to follow.	
Interaction Design	**5 to >3.75 Pts Excellent** Interactive features are clearly described and	**3.75 to >3.25 Pts Good** Interactive features are described and generally	**3.25 to >2.75 Pts Satisfactory** Interactive features are identified but justifications are	**2.75 to >2.25 Pts Limited** Some interactive elements are mentioned, but descriptions are	**2.25 to >0 Pts Unsatisfactory** Interactive features are missing, unclear, or	5 pts

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	well justified in relation to the visualisation's purpose and user needs. The role of interactions such as filtering, zooming, tooltips, animation, or highlighting is clearly explained, showing how they improve usability, focus attention, or support exploration without overwhelming the user. Interactions are purposeful, intuitive, and well aligned with the key questions users are expected to answer. The design demonstrates a strong understanding of how interaction can support (or hinder), analytical exploration and explanatory storytelling.	appropriate for the visualisation goals. Their role in supporting user experience and understanding is explained, though justifications may be brief. Interactions support exploration and comprehension as part of story telling.	at a high level or generic. The interaction table includes basic interaction details. The connection between interactions and user goals is present but weakly justified. Some interactions may appear unnecessary or poorly motivated.	unclear, incomplete, or poorly justified. Expected user behaviour or system responses are vague or inconsistent. Interactions show limited consideration of usability or user intent. Users are provided with fully interactive visualisations (e.g., the ability to filter for every variable) without considering usability of interactivity.	inappropriate. The interaction table is missing or does not meaningfully describe user interactions. There is little evidence of planning for user behaviour or interactive support for understanding the data. Content is generic, vague, or contains few concrete examples. Lacks logical flow, content is difficult to follow.	
Iteration and Validation	**5 to >3.75 Pts Excellent** The design process clearly demonstrates an iterative	**3.75 to >3.25 Pts Good** A usability evaluation is conducted and described, with	**3.25 to >2.75 Pts Satisfactory** Some usability testing and refinement are described, but	**2.75 to >2.25 Pts Limited** Usability evaluation is basic, poorly described, or	**2.25 to >0 Pts Unsatisfactory** At least one of the following: There is little or no evidence of	5 pts

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	approach to testing and refinement, with multiple rounds of improvement informed by feedback from peers, users, or instructors. Evaluation method and feedback is clearly documented, and changes made in response are explicitly explained and justified. Before-and-after comparisons using screenshots or sketches effectively illustrate how the design evolved. Adjustments made due to technical or programming constraints are clearly discussed, including how design intent was preserved or adapted. Accessibility is formally assessed and demonstrated. A usability evaluation is conducted using appropriate methods, with	clear findings that inform at least some design improvements. Usability testing method and results are well documented. Clear examples of changes made are documented. Before-and-after comparisons are provided and generally support the discussion. Technical or programming constraints are acknowledged, with reasonable explanation of resulting design changes. Accessibility considerations are addressed, though not always in depth.	iteration is limited or unevenly documented. Feedback is mentioned, though responses to feedback may be brief or lack clear justification. Before-and-after comparisons are provided but not well documented or offer limited insight into decision-making. Technical or accessibility considerations are acknowledged at a surface level. A basic usability evaluation is conducted, but methods, findings, or outcomes are described with limited detail.	weakly connected to design improvements. Limited evidence of iteration or refinement is provided. Feedback is minimal, vague, or not clearly linked to design changes. Before-and-after comparisons are incomplete or difficult to interpret. Discussion of technical constraints or accessibility is minimal.	usability testing. Feedback is absent, and changes to the design are not documented or justified. Accessibility and usability considerations are not meaningfully addressed. Content is generic, vague, or contains few concrete examples. Lacks logical flow, content is very difficult to follow.	

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	clearly defined tasks, participants, and evaluation criteria. Findings are thoughtfully analysed and used to inform further refinements, demonstrating a strong understanding of usability principles.					
Implementation 1: Coding Practice	**5 to >3.75 Pts Excellent** The website functions as expected. The code demonstrates consistently high-quality programming practice. The overall structure is clear and modular, with a logical separation of concerns (e.g. data handling, visualisation logic, interaction, and presentation). D3 is used appropriately and effectively to bind data, construct visual elements, and manage updates or interactions. Code is readable and well organised,	**3.75 to >3.25 Pts Good** The website functions as expected. The code is well structured and demonstrates sound programming practice. D3 is used appropriately for data binding, chart construction, and interaction. Variable naming and organisation are generally clear, and the code is readable with minimal redundancy. Some modularisation is evident, though separation of concerns may not be fully consistent. Data handling is mostly robust, with minor issues	**3.25 to >2.75 Pts Satisfactory** The website functions as expected with minor issues. Code is mostly well structured and modularised. Consistent informative commenting is present. Data handling and state management work for expected cases. Accessibility features may be present.	**2.75 to >2.25 Pts Limited** Website functions mostly as expected, but somethings may not work properly (e.g., screen resizing, elements not properly visible, filter updates, tool tips etc). Some modularisation is apparent but does not follow conventions taught in unit. Commenting is poor or inconsistent.	**2.25 to >0 Pts Unsatisfactory** One of the following: ### D3 is not used to code visualisations in website (or appears to be used via another package). ### The website does not display correctly or does not open at all. Code is poorly modularised (e.g., data loading, interactions, chart building, styling and constants are not clearly separated). The code is not appropriately formatted. ### The code is not commented. The use of AI	5 pts

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	with meaningful variable and function names and clear commenting that identifies use of GenAI code. Data loading, transformation, and state management are handled robustly. Error handling and edge cases (e.g. missing or unexpected data) are thoughtfully managed. The code supports accessibility and usability where relevant (e.g. semantic structure, ARIA attributes, keyboard interaction, readable text) and is clearly identified in comments. Overall, the implementation reflects a strong understanding of web and D3 programming principles.	or omissions. Accessibility features are present and clearly identified in comments.			code is not clearly identified. ###	
Implementation 2: Complexity	**5 to >3.75 Pts Excellent** The project demonstrates a high level of technical complexity that goes well	**3.75 to >3.25 Pts Good** The project demonstrates a solid level of technical complexity, including some	**3.25 to >2.75 Pts Satisfactory** The project demonstrates moderate technical complexity, largely based on	**2.75 to >2.25 Pts Limited** The project demonstrates minimal technical complexity. Visualisations and web	**2.25 to >0 Pts Unsatisfactory** The project shows insufficient technical complexity or incomplete	5 pts

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	beyond standard examples covered in labs or tutorials. The visualisation design includes non-standard or advanced visualisation techniques that are thoughtfully selected and effectively implemented. Multiple datasets are integrated in a meaningful way, with clear justification for how they enhance analysis or interpretation. Web programming techniques including css and JavaScript are used competently to support interaction, responsiveness, and overall functionality. Technical choices are appropriate, well executed, and clearly aligned with the project's analytical and communication goals, demonstrating a strong level of technical	elements not directly covered in labs. The visualisation design extends beyond basic chart types, and data integration or advanced web features are used appropriately. Multiple datasets may be combined, or non-trivial data processing is undertaken. Web programming supports interaction and structure effectively, though some aspects may rely on known patterns or provided examples. Technical ambition is evident and mostly well realised.	techniques introduced in labs or taught examples, with limited extension. Visualisations are competently implemented but rely primarily on standard approaches. Data integration and web programming are present but simple or narrowly scoped. The implementation meets expectations but shows limited technical innovation or challenge.	programming are basic and closely follow lab materials with little adaptation. Data usage is simple, and there is little evidence of extension, integration, or technical challenge beyond required minimums.	implementation. Visualisation techniques, data handling, or web programming are missing, incorrect, or largely non-functional.	

Criteria	Ratings					Pts
	independence and problem-solving.					
Implementation 2: Storytelling	**5 to >3.75 Pts** **Excellent** The project presents a clear, coherent, and well-structured data story that guides the user through the analysis in a logical and meaningful way. The narrative explicitly connects the analytical questions, data evidence, and visualisations to form a unified whole. Visualisations are sequenced and framed to support key insights, with clear transitions that help users understand how observations build on one another. The story balances explanation and exploration effectively, highlighting important findings while allowing users to engage with the data. Claims and interpretations are consistently	**3.75 to >3.25 Pts** **Good** The project presents a clear narrative that communicates key insights drawn from the data. The structure generally supports the analytical story, and visualisations are mostly well sequenced and relevant to the stated purpose. Connections between questions, data, and conclusions are evident, though some transitions or explanations may be underdeveloped. The narrative supports understanding but may lack depth or cohesion in parts.	**3.25 to >2.75 Pts** **Satisfactory** The project communicates some insights from the data, but the story is loosely structured or uneven. Visualisations are relevant but may feel disconnected or insufficiently framed within a clear narrative. The data story is present but relies on the reader to infer connections between visuals and insights. Interpretations may be descriptive rather than analytical.	**2.75 to >2.25 Pts** **Limited** The project presents visualisations with minimal narrative structure. The overall story is unclear, fragmented, or overly focused on individual charts rather than a coherent message. Users are given options to explore the data, but there is limited effort to guide the user through an analysis to answer specific questions identified in the introduction.	**2.25 to >0 Pts** **Unsatisfactory** The project does not present a meaningful data story. Visualisations lack narrative framing, and there is little or no connection between the data, insights, and stated purpose of the project.	5 pts

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	grounded in data, and the narrative demonstrates a strong understanding of the broader context and implications of the findings.					
References	**3 to >2.25 Pts Excellent** Sources are highly relevant, credible, and appropriate to the project context, including academic literature, authoritative government or industry sources, and relevant design or visualisation research where applicable. The reference section is complete, accurate, and consistently formatted using APA7 style. Sources demonstrate strong engagement with the problem domain and relevant theory or best practice.	**2.25 to >1.95 Pts Good** Sources are relevant and credible, supporting key aspects of the project such as data context, design decisions, or usability considerations. The reference section is mostly complete and correctly formatted in APA7, with minor inconsistencies. Sources demonstrate sound understanding of the domain, though engagement with literature may be limited in breadth or depth.	**1.95 to >1.65 Pts Satisfactory** Some relevant sources are included, but coverage may be narrow, uneven, or focused mainly on general web resources. References support parts of the project but are not consistently integrated with analysis or design justification. Formatting uses APA7 but may contain errors or inconsistencies.	**1.65 to >1.35 Pts Limited** Few sources are included, and they provide limited support for the project's claims or decisions. The reference section may be incomplete, poorly formatted in APA7, or rely heavily on non-authoritative sources. Referencing shows minimal engagement with relevant literature or context.	**1.35 to >0 Pts Unsatisfactory** References are missing, largely irrelevant, or inappropriate. Sources do not meaningfully support the project, or the reference section is absent or not formatted according to APA7. Use of fake references will result in OVERALL FAIL in assessment.	3 pts
						Total points: 53